

Crop Storage — Module Evolution

Christelle Alvine Dehoumon · Infineon / BIT Excellence Program 2026 · Sustainable dried-mango supply chain
(Burkina Faso → Germany)

Timeline (CW 07/24/2026 → CW 09/11/2026)

Week	Focus	Main achievements
CW 07/24	Concept & requirements	Initial concept of a digital Crop Storage solution (Digi Mango); reviewed the five official functions (inventory, temperature & humidity monitoring, FIFO, loss & waste tracking, capacity planning); studied last year's mango storage work.
CW 07/31	Data & first mockup	Analyzed the 2025 Excel data and peer homepages; defined storage data requirements; designed the first UI mockup (Digi Mango) with real-time capacity usage; integrated Hans & Abdo feedback (storage capacity status tracking).
CW 08/07	Rebrand & planning	Rebranded Digi Mango into CropCore; researched realistic storage data; organized the task breakdown on the Miro board.
CW 08/14	Design consolidation	Added a new scenario to the dashboard; created a unique farm logo for a consistent identity; restructured the Miro board for clarity and tracking.
CW 08/21	Scenario Analysis v2	Extended the What-if Scenario Analysis (operating costs, net benefits, break-even timing, five-year cumulative view); improved dashboard navigation; refined tasks, milestones and dependencies on the Miro board.
CW 08/28	Design system & backend planning	Applied the shared typography and circular logo standards; clarified graph axes, FCFA units and data assumptions (Hans feedback); reviewed the shared repository and outlined the initial Crop Storage API and PostgreSQL data model.
CW 09/04	Environment & first vertical slice	Set up and verified the environment (Docker, Docker Compose v2, Git); cloned the monorepo and switched to feature/cropstorage/init; launched the full stack (PostgreSQL 17, Spring Boot 4.1, React/nginx); implemented the first end-to-end Batch flow (POST/GET /api/batches).
CW 09/11	Storage zones & sensors	Completed the StorageZone module and linked batches to zones (foreign key verified in PostgreSQL); implemented the Sensor module (entity, repository, service, controller) with POST/GET /api/sensors; created the SensorReading entity linked to sensors; updated the Miro board.

Current backend status

Entity / feature	Status	Details
Batch	Done	Entity with validation, repository, service, controller. POST/GET /api/batches. Persists to PostgreSQL.
StorageZone	Done	Entity, repository, service, controller. Linked to Batch (storage_zone_id).
Sensor	Done	Entity, repository, service, controller. POST/GET /api/sensors. Linked to StorageZone.
SensorReading	In progress	Entity created and linked to Sensor; repository/service/controller next.

StockMovement, LossRecord, Alert	Planned	Next entities after SensorReading.
FIFO, capacity planning, scenarios	Planned	Business logic to add after the data model.
React dashboard	Planned	Frontend not started yet (React 19, Vite, Tailwind 4, Recharts).

Technical stack

- **Backend:** Java 21, Spring Boot 4.1, Maven, Spring Data JPA, PostgreSQL 17, Validation, Actuator, Springdoc OpenAPI (Swagger), Lombok.
- **Frontend:** React 19, Vite, Tailwind CSS 4, Lucide React, Recharts (not started yet).
- **Infrastructure:** Docker Compose — PostgreSQL, Spring Boot backend, React/nginx frontend.
- **Reference:** interactive mockup published at <https://christelle-dehoumon.github.io/CropCore-Dashboard-Mockup/>

Next steps

- Complete the SensorReading module (repository, service, controller).
- Implement stock movements (IN / OUT / TRANSFER / ADJUSTMENT) and the FIFO logic.
- Add loss and waste tracking, alerts, capacity planning and scenario endpoints.
- Start the React dashboard on top of the REST API.
- Confirm open business questions with Anar (batch examples, thresholds, FIFO rule, product value, sensor format).